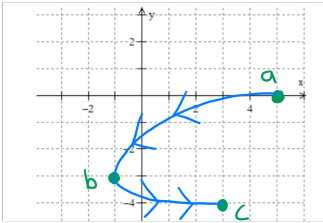


Polar and Cartesian Coordinates

given: $x = 3 - t^3 + 3t$, $y = t^2 - 4$ $-2 \leq t \leq 0$



include the direction of where the points are leading

t	x	y	
2	5	0	a (5, 0)
-1	1	-3	b (1, -3)
0	3	-4	c (3, -4)

- the t values consist of x and y
You are given an interval all the values in that interval consist of t-values to graph this you will set the t-value as the input for the x and y-equation

example:

$$x = 3 - (-2)^3 + 3(-2) = 5$$

$$y = (-2)^2 - 4 = 0$$

these are for the t-value $t = -2$

Finding the $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2t}{-3t+3}$$

$$x = 3 - t^3 + 3t \quad y = t^2 - 4$$

$$dx/dt = -3t + 3 \quad dy/dt = 2t$$

- take the derivative of y-equation and divide it by the derivative of x-equation

Finding the $\frac{d^2y}{dx^2}$

$$\frac{\text{derivative of } \left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{12t-6}{(-3t+3)^2}$$

- take the derivative of dy/dx and divide it by dx/dt

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

$$f = 2t$$

$$g = -3t + 3$$

$$\frac{df}{dx} = 2$$

$$\frac{dg}{dx} = -3$$

$$\frac{(-2) \cdot (-3t+3) - (2t) \cdot (-3)}{(-3t+3)^2} \Rightarrow \frac{12t-6}{(-3t+3)^2}$$

Find the equation of the line to the curve at the point $t = -2$

$$x(2) = 3 - (2)^3 + 3(2) = 5$$

$$(5, 0) \quad m = -\frac{4}{3}$$

$$y - y_1 = m(x - x_1)$$

$$y(2) = (2)^2 - 4 = 0$$

$$y - 0 = -\frac{4}{3}(x - 5)$$

$$m = \frac{dy}{dx} = \frac{2t}{-3t+3} = \frac{2(2)}{-3(2)+3} = -\frac{4}{3}$$

$$y = -\frac{4}{3}x + \frac{20}{3}$$

Vertical and horizontal tangent

$$\text{horizontal} \Rightarrow \frac{dy}{dx} = 0 \quad \text{or} \quad \frac{dy}{dt} = 0$$

$$\text{Vertical} \Rightarrow \frac{dx}{dt} = 0$$

Set the wanted value
either Vertical or horizontal
to zero and Solve for
 x

Setting up the integral that represents the length of the curve

$$\int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

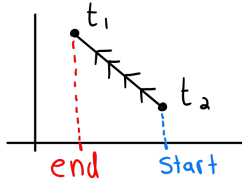
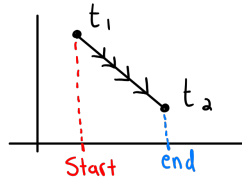
- this goes from the absolute smallest to the absolute large for the limit of the integral

Area - Curve to x-axis

$$\int_{t_1}^{t_2} y(t) \cdot \frac{dx}{dt} \cdot dt$$

- What matter is where it starts and where it ends

- the integral limits must be in "t" values



$$\int_{t_1}^{t_2} y(t) \cdot \frac{dx}{dt} \cdot dt$$

- it starts at t_1 so t_1 goes to the bottom and it ends at t_2

$$\int_{t_2}^{t_1} y(t) \cdot \frac{dx}{dt} \cdot dt$$

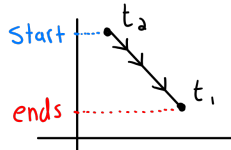
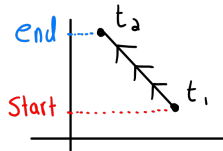
- it starts at t_2 so it goes to the bottom and it ends at t_1

area-curve to y-axis

$$\int_{t_1}^{t_2} x(t) \cdot \frac{dy}{dt} \cdot dt$$

- What matter is where it start and where it end

- the integral limits must be "t" values



$$\int_{t_1}^{t_2} x(t) \cdot \frac{dy}{dt} \cdot dt$$

- it begins from t_1 and ends at t_2

$$\int_{t_2}^{t_1} x(t) \cdot \frac{dy}{dt} \cdot dt$$

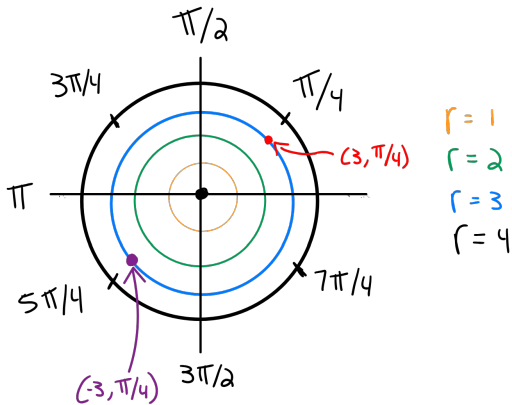
- it begins at t_2 and ends at t_1

Finding the speed at time "t"

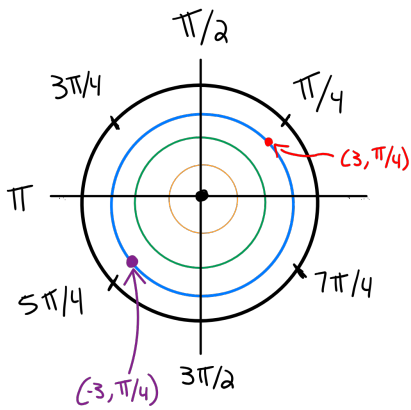
$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

- after you found $\frac{dx}{dt}$ and the $\frac{dy}{dx}$ at the given t-value Plug it into the square root
- there is no integration you will find the derivation

Polar Coordinates



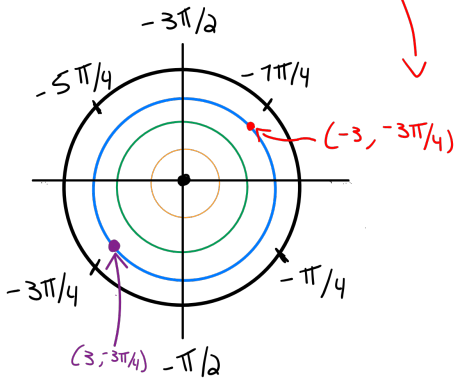
- each circle represents a value of r
- for a regular graph we have (x, y) for this it goes (r, θ) before you began looking for the point it would make simpler to begin finding " θ " or radians
- if we wanted to plot $(3, \pi/4)$ we began by finding $\pi/4$ we found $\pi/4$ and we know that $r=3$ is the blue line so that is where we place $(3, \pi/4)$
- what about plotting $(-3, \pi/4)$ we would ignore the negative and find $\pi/4$ and we know that $r=3$ is the blue line that point is $(3, \pi/4)$ to make this negative we take it right across that will be $(-3, \pi/4)$ as soon as you move the point across it becomes negative



$$\left. \begin{matrix} (3, \frac{5\pi}{4}) \\ (3, -\frac{3\pi}{4}) \end{matrix} \right\}$$

- these Point will also get you to the Same Point as $(-3, \pi/4)$ this is because each Point in Polar coordinates can represent many different coordinates or "different routes Will get you to the Same Place" routes Meaning $(3, \frac{5\pi}{4})$ and $(-3, \pi/4)$ get you to the Same Point (infering the $(-3, \pi/4)$ and $(3, \frac{5\pi}{4})$ are routes)

- that Point would also be $(3, -7\pi/4)$

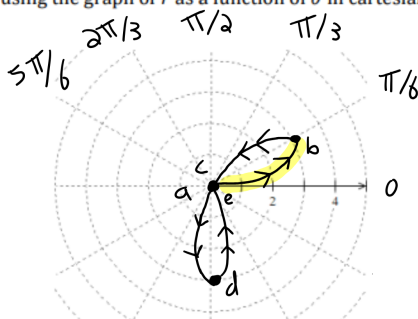
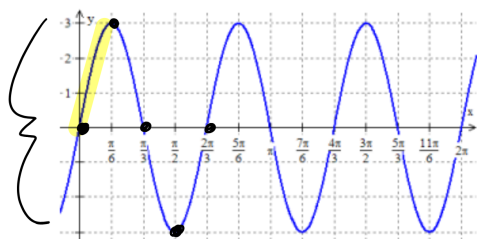


- this has negative radian but it still works the same Just with negative instead of Positive which does not Change a whole lot lets Plot $(3, -3\pi/4)$ We take the same exact Steps as we did with the positive radians What if we Wanted to Plot $(-3, 3\pi/4)$ we do the Same thing as we did for the positive radians the Point is they the Same Just With negative radians

- $\left. \begin{matrix} (-3, \pi/4) \\ (3, 5\pi/4) \\ (-3, -7\pi/4) \end{matrix} \right\}$ all of these Point will get you to the Same Point as $(3, -3\pi/4)$

8. Sketch the curve with the given polar equation by using the graph of r as a function of θ in cartesian coordinates for $0 \leq \theta \leq 2\pi/3$

r-Values



a and b are within the $0 \leq \theta \leq \pi/6$ region

- look at the highlighted which goes from 0 to $\pi/6$ these are two points (0,0) and $(3, \pi/6)$ which is point a and b they must be within that region

- began by making a plot from the graph you are given a interval from $0 \leq \theta \leq \frac{2\pi}{3}$ stay within that interval

- a (0,0)
- b $(3, \pi/6)$
- c $(0, \pi/3)$
- d $(-3, \pi/2)$
- e $(0, 2\pi/3)$

- these are (r, θ)

Finding Slope of the tangent line to the curve $r = \sin \theta$ at $\theta = \pi/4$

- $x = r \cos \theta$
- $y = r \sin \theta$

- take the given "r" value and plug it into each of the equations. You will take the derivative of x and y then divide the y-derivative by the x-derivative or after you take each derivative you can follow the equation below after that you plug in the "theta" into that

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

Converting From Cartesian to Polar

Cartesian = (x, y)

Polar = (r, θ)

$$\bullet x^2 + y^2 = r^2$$

$$\bullet \tan \theta = \frac{y}{x}$$

} For converting
Cartesian to
Polar

ex: $(-4, 4)$

$$x^2 + y^2 = r^2 \Rightarrow \sqrt{(-4)^2 + (4)^2} = 4\sqrt{2}$$

$$\tan \theta = \frac{y}{x} \Rightarrow \tan \theta = \frac{-4}{4} = -1 \quad \bullet \text{Find Where } \tan \theta = -1$$

$$\theta = 3\pi/4$$

$$(4\sqrt{2}, 3\pi/4)$$

Converting the Following Polar Coordinates to Cartesian

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned}$$

} these are the equations that
will help convert from polar to
Cartesian

ex: $(3, -\frac{5\pi}{4})$

• the 3 is the "r-value"

$$x = r \cos \theta \Rightarrow \cos(-\frac{5\pi}{4}) = -\frac{\sqrt{2}}{2} \Rightarrow 3(-\frac{\sqrt{2}}{2}) = -\frac{3\sqrt{2}}{2}$$

$$y = r \sin \theta \Rightarrow \sin(-\frac{5\pi}{4}) = \frac{\sqrt{2}}{2} \Rightarrow 3(\frac{\sqrt{2}}{2}) = \frac{3\sqrt{2}}{2}$$

$$\left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$$

Converting Polar to Cartesian equation

- $x^2 + y^2 = r^2$
- $\tan \theta = \frac{y}{x}$
- $x = r \cos \theta$
- $y = r \sin \theta$
- equation that will help you convert Polar to Cartesian or vice versa

$$\bullet \cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

$$\bullet \sin(2\theta) = 2 \sin \theta \cos \theta$$

- these are identities that may help with replacing some stuff in an equation to make stuff easier

ex:

$$r = 3 \cos \theta$$

$$\bullet x = r \cos \theta$$

$$r = 3 \left(\frac{x}{r} \right)$$

$$\frac{x}{r} = \cos \theta$$

$$r^2 = 3x$$

$$\bullet x^2 + y^2 = r^2$$

$$x^2 + y^2 = 3x$$

$$\boxed{x^2 + y^2 = 3x}$$

- We know this is a Polar equation because there is a trig function, also there is a " θ " not a " x " it also says " r " equals

Convert the Cartesian to Polar equation

$$x^2 + y^2 = 4x$$

$$\bullet x^2 + y^2 = r^2$$

$$r^2 = 4r \cos \theta$$

$$\bullet x = r \cos \theta$$

$$\boxed{r = 4 \cos \theta}$$

$$y = -x$$

$$\bullet y = r \sin \theta$$

$$r \sin \theta = -r \cos \theta$$

$$\bullet x = r \cos \theta$$

$$\sin \theta = -\cos \theta$$

$$\frac{\sin \theta}{\cos \theta} = -1 \Rightarrow \tan \theta = -1$$

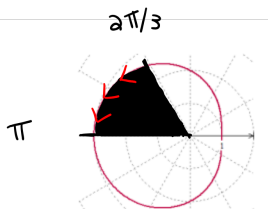
$$\theta = \tan^{-1}(-1)$$

$$\boxed{\theta = \frac{7\pi}{4}}$$

- you must isolate the " r " or " θ " completely

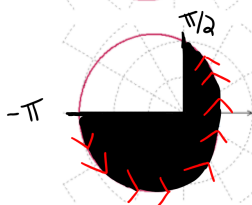
given $r = 2 - \cos \theta$ Shade the area represented by the given integral

a) $\int_{2\pi/3}^{\pi} \frac{1}{2} r^2 d\theta$



- always goes counter-clockwise
also these go from small to large

b) $\int_{-\pi}^{\pi/2} \frac{1}{2} r^2 d\theta$



- it must go from smallest to the largest in the counter-clockwise direction

this the equation you would use when calculating area either for a shape or when you are given the r -equation and a interval

- the red lines represent the direction of there movement

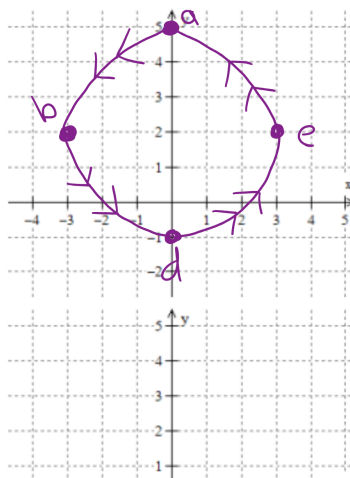
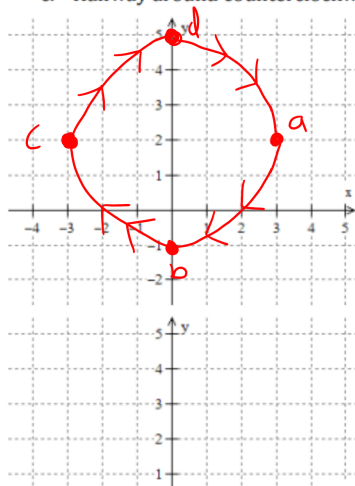
$$\int_{\theta_1}^{\theta_2} \frac{1}{2} r^2 \cdot d\theta$$

Finding the Point of intersection

- to Find the Point of intersection between two " r -equations" you must set the two equation equal to each other and solve for " θ " you may have more then one answer to get " r -value" For each one of the answer you can plug the answer you got for the question into either one of the given " r -equation" doesn't matter which one you plug the answer into if there are many different answers you do this For each one this is what the point is supposed to looklike (r, θ)

Find parametric equations for the path of a particle that moves along the circle $x^2 + (y - 2)^2 = 9$ in the manner described.

- Once around clockwise, starting at (3, 2).
- Three times around counterclockwise, starting at (0, 5).
- Halfway around counterclockwise, starting at (0, -1).



• look at the Next Page

- began by Finding the Parametric equation

$$x^2 + (y - 2)^2 = 9 \Rightarrow \frac{x^2}{9} + \frac{(y - 2)^2}{9} = 1$$

$$\frac{x^2}{9} = \cos^2 x \Rightarrow \sqrt{x^2} = \sqrt{9 \cos^2 x} \Rightarrow x = 3 \cos x$$

$$\frac{(y - 2)^2}{9} = \sin^2 x \Rightarrow \sqrt{(y - 2)^2} = \sqrt{9 \sin^2 x} = y - 2 = 3 \sin x + 2$$

t	x	y	
0	3	2	a(3, 2)
$\pi/2$	0	-1	b(0, -1)
π	-3	2	c(-3, 2)
$3\pi/2$	0	5	d(0, 5)
2π	3	2	e(3, 2)

- in question a it wanted us to start from (3, 2) but we got (0, 5) so we don't use it and go to the next number but if you don't use it you must add another number to take its place to the end we added $\pi/2$ to 2π giving us $5\pi/2$

$$\sin^2 x + \cos^2 x = 1$$

- this identities will help you convert

- by setting $x = \cos x$ and $y = \sin x$ it goes clockwise

- your interval must go from $0 \leq t \leq 2\pi$ or $-2\pi \leq t \leq 0$ this is for clockwise or counterclockwise

- this goes by $\pi/2$

- to get it to go Counterclockwise we need to set $x = \sin x$ and $y = \cos x$

$$x^2 + (y-2)^2 = 9 \Rightarrow \frac{x^2}{9} + \frac{(y-2)^2}{9} = 1$$

$$\frac{x^2}{9} = \sin^2 x \Rightarrow \sqrt{x^2} = \sqrt{9 \sin^2 x} = x = 3 \sin x$$

$$\frac{(y-2)^2}{9} = \cos^2 x \Rightarrow \sqrt{(y-2)^2} = \sqrt{9 \cos^2 x} \Rightarrow y = 3 \cos x + 2$$

- we are placing "y" as "cos x" and "x" as "sin x" to get the graph to go the opposite way (Counterclockwise)

t	x	y	
0	3	2	
$\pi/2$	0	5	a (0, 5)
π	-3	2	b (-3, 2)
$3\pi/2$	0	-1	c (0, -1)
2π	3	2	d (3, 2)
$5\pi/2$	0	5	e (0, 5)